MATH 231 – Linear Algebra I

Computational Projects & Application Demos

Fall 2026 ▪ working draft — not finalized

This document is not finalized. It is a menu of candidates, not a list of assignments. Nothing here is part of the syllabus, and no item should be treated as assigned unless it has been officially announced. Where this document and an official course document disagree, the official document is correct.

Assessment mapping (draft)

Counts here are stale: this sketch shows four projects and ten homework assignments, while the course carries **three projects and five homework assignments**. Retained for the unit-coverage mapping only; the root `README.md` holds the authoritative weights.

Assessment	Units covered	Suggested focus
Exam 1	1–6	vector spaces, norms, complex, matrices, calculus, conditioning
Exam 2	7–10	systems, maps, subspaces, least squares
Final Exam	1–12 (weighted to 11–12)	eigen/SVD, optimization, synthesis
Project 1	after 2	#1 K-means image color segmentation
Project 2	after 11	#5 Eigencats via PCA
Project 3	after 12	#7 Mini-ML: OLS + SVM + KNN with validation
Project 4	after 12	#8 Signal trilateration: Gauss–Newton / LM
Lecture demos	10–12	#2 SVD+DCT, #3 LSA, #4 collab. filtering, #6 PageRank, #9 embeddings
Homework	one per unit (Units 1–12, some combined)	mixed hand computation + Python verification

The project menu

This is a *menu*, not an assignment quota: every item below is documented, but only a small subset is assigned to students as an implementation; the complement are instructor-led demos that still put the application in front of the class. The list is open — further items may be appended as #10, #11, … without disturbing the ordering.

Outline additions these items introduced: DCT / orthogonal transforms (Unit 10); Perron–Frobenius theorem and stochastic matrices (Unit 11); TF–IDF term–document weighting (Unit 11); masked-loss / ALS factorization and train/validation/test model selection (Unit 12).

#	Project	Role	Prereq	LA machinery
1	K-means color segmentation	Assigned	U2	K-means, image vectorization, ℓ_2 distance
2	SVD + DCT image compression	Demo	U11	SVD, Eckart–Young, DCT orthonormal basis
3	LSA file-system search	Demo	U11	truncated SVD, TF–IDF, cosine similarity
4	Collaborative filtering	Demo	U12	masked low-rank factorization, ALS/SGD
5	Eigencats (PCA)	Assigned	U11	PCA, covariance eigenbasis / SVD
6	PageRank	Demo	U11	power method, stochastic matrix, Perron–Frobenius
7	Mini-ML (OLS/SVM/KNN)	Assigned	U12	OLS, SVM, KNN, train/val/test
8	Signal trilateration	Assigned	U12	Gauss–Newton, Levenberg–Marquardt, Jacobian
9	Semantic embedding geometry	Demo	U11	cosine similarity, KNN, PCA, vector arithmetic

All implementations are in **Python**.

Candidate assigned implementations

#1 – K-means image color segmentation (after Unit 2)

Cluster an image's pixels in RGB (or CIELAB) space into k colors and replace each pixel by its cluster centroid; display the quantized/segmented image as a function of k. *LA focus:* ℓ_2 distance, centroids as means, Lloyd's iteration and its convergence. Python (NumPy/Pillow).

#5 – Eigencats via PCA (after Unit 11)

Assemble a data matrix of aligned cat (or face) images, mean-center, and compute principal components via the SVD; visualize the top "eigencats," reconstruct images from r components, and plot reconstruction error and variance-explained vs. r. *LA focus:* covariance eigenstructure, SVD, low-rank reconstruction. Python.

#7 – Mini-ML with validation (after Unit 12)

Three sub-tasks on suitable datasets: OLS for a regression problem, an SVM for a classification problem, and KNN for a multiclass problem; use a train/validation/test split, report generalization metrics, and justify model choice. *LA focus:* normal equations vs. QR (OLS), margin/kernels (SVM), distance metrics (KNN), validation methodology. Python (hand-implement the cores; library baselines allowed). *Depends on topics currently listed as not guaranteed — see `MATH231_tentative_topics`.*

#8 – Signal trilateration (after Unit 12)

From noisy range measurements to known anchors, estimate an emitter's position by nonlinear least squares; implement Gauss–Newton and Levenberg–Marquardt and compare convergence/robustness vs. initialization and noise. *LA focus:* residual Jacobian, GN normal-equation step, LM damping, conditioning. Python.

Candidate lecture demos

#2 – SVD + DCT image compression (Unit 11)

Sweep the rank r of an image's SVD and watch quality vs. storage; then show the 8×8 block DCT that JPEG actually uses, connecting a *fixed* orthonormal basis to the data-adaptive SVD basis. Motivates Eckart–Young.

#3 – Latent Semantic Analysis search (Unit 11)

Build a small term–document matrix over a mock file system, TF–IDF weight it, truncate its SVD, and answer queries by cosine similarity in the reduced "concept" space; contrast with exact keyword matching.

#4 – Collaborative filtering (Unit 12)

On a toy ratings matrix with missing entries, fit a low-rank factorization by ALS/SGD over *observed* entries and predict held-out ratings; frames recommendation as matrix completion.

#6 – PageRank (Unit 11)

Model a small web graph as a stochastic matrix, run the power method, and watch the ranking vector converge to the dominant eigenvector; connect to Perron–Frobenius and the damping factor.

#9 – Semantic embedding geometry (Unit 11, static)

With pretrained word vectors: nearest neighbors under cosine similarity, analogy arithmetic (king $-$ man $+$ woman $\approx$ queen), PCA to 2-D for visualization, and clustering — purely linear algebra, no model training.